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Author signature: *Will Meany*

ME3330 – Heat Exchangers Technical Summary

William Meany
Department of Mechanical Engineering, Clemson University, Clemson, SC 29634, USA
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Deliverable 1:

Strain gauges are an important sensor for data collection and safety. Human eyesight is only able to observe changes in length and displacement at a certain scale. This is where the strain gauge is able to supplement the visual inspections with data showing change in length. Visual deformations such as cracking are visible in some materials, but defects can present in other unpredictable ways. Metal properties are very well studied, and all metal behaving materials have an elastic region of their stress strain curve. This is the region where the metal will return to its original unstressed state, if the stress is removed. A strain gauge can monitor the strain resulting from the stress, and ensure metals are operating within their elastic region.

Strain gauges are applied to the surface of interest with a bonding agent such as epoxy or glue. As the surface changes in length, the strain gauge does as well. A strain gauge is composed of many wires running parallel to each other in the direction of strain. They can be seen below in FIG.1. as the resistive foil.

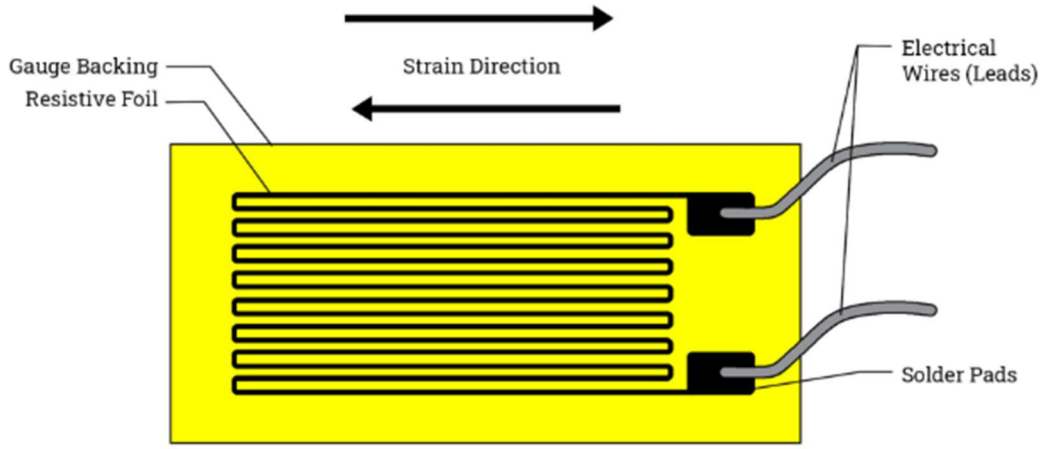


FIGURE 1: Strain Gauge layout and components.

These wires run a small amount of electrical current through them. When the area of interest is deformed through strain, the wires will experience the same strain. This leads to a change in the cross sectional area of the wire, which causes electrical resistance. The equation governing this phenomenon is equation 1.

$$R = \frac{\rho \cdot L}{A} \tag{1}$$

The length of the wire (L), and the resistivity of the material (ρ), are both constant, therefore Equation 1 states that a decrease in cross sectional area leads to an increase in the resistance. This increase in the resistance is detected and converted to a measurement how much the sensor has gained in length. This can

be used to calculate strain which is the relationship between the change in length and the original length.

$$\varepsilon = \frac{\Delta L}{L} \tag{2}$$

This relationship between the resistance and strain can also be used to calculate a factor known as gauge factor (GF). This is a means to compare different types of strain gauges, because a high gauge factor refers to a high sensitivity of the gauge. The equation to calculate gauge factor is below.

$$GF = \frac{\Delta R/R}{\varepsilon} \tag{3}$$

The resistance recorded from the sensor is not immediately converted into strain. A quarter wheatstone circuit is applied to ensure that the small changes in resistance are able to be measured. This circuit involves four resistors with the strain gauge being one of the resistors. A diagram of the wheatstone circuit used in the strain gauge is below in FIG. 2.

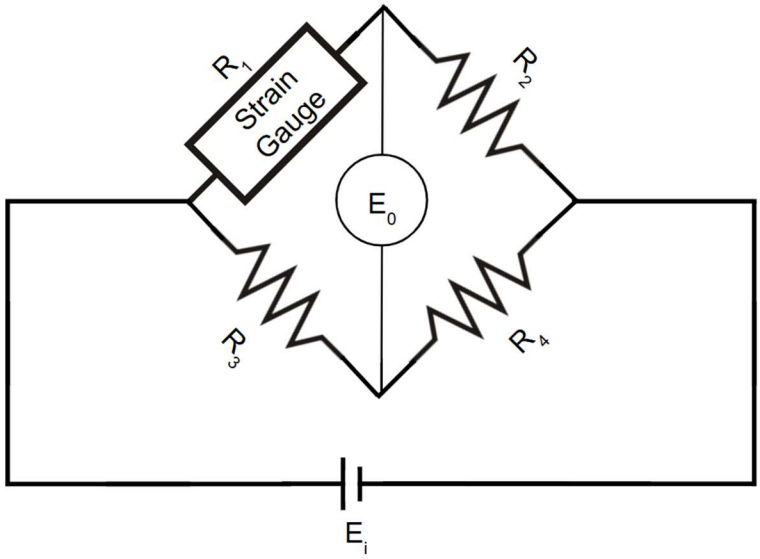


FIGURE 2: Quarter wheatstone circuit diagram

The quarter wheatstone circuit balances the voltage on both the top and bottom, and when in equilibrium, there is zero voltage output through E_0. When one of the resistances changes a measurable amount of voltage is detected at the output. This value can be recorded and used to calculate the strain. It is non-essential, but it improves accuracy, and accounts for changes in the environment. Temperature changes would normally affect resistance, but due to all resistors balancing each other out, and being a part of the strain gauge, the temperature would affect all of the resistors effectively preventing it from skewing the data.

The voltage output can be manipulated through the known values of R_1, R_2, R_3, R_4, and E_i. The resistance of $R_1$is defined in equation 4 as the sum of the initial resistance of the strain gauge, and the change in resistance.

$$R_1 = R_i + \Delta R \tag{4}$$

The resistances are connected through the following relationship in equation 5.

$$R_i = R_2 = R_3 = R_4 = R \tag{5}$$

Through circuit analysis, the following equation can be derived.

$$\frac{\Delta \mathrm{R}}{R} = \frac{4\Delta E_0}{E_i} \tag{6}$$

Thus, plugging the result from equation 6 into equation 3, the relationship between voltage, strain, and gauge factor can be described by equation 7.

$$\varepsilon = \frac{4\Delta E_0}{E_i GF} \tag{7}$$

With the conceptual understanding of how a strain gauge works, and how it converts a small change in length, to a small change in resistance, to a large change in volage, the experiment can begin. The strain gauge was attached to the top of the aluminum 6061 beam. The experimental setup can be seen below in FIG 3., along with the dimensions which are used later for stress calculations.

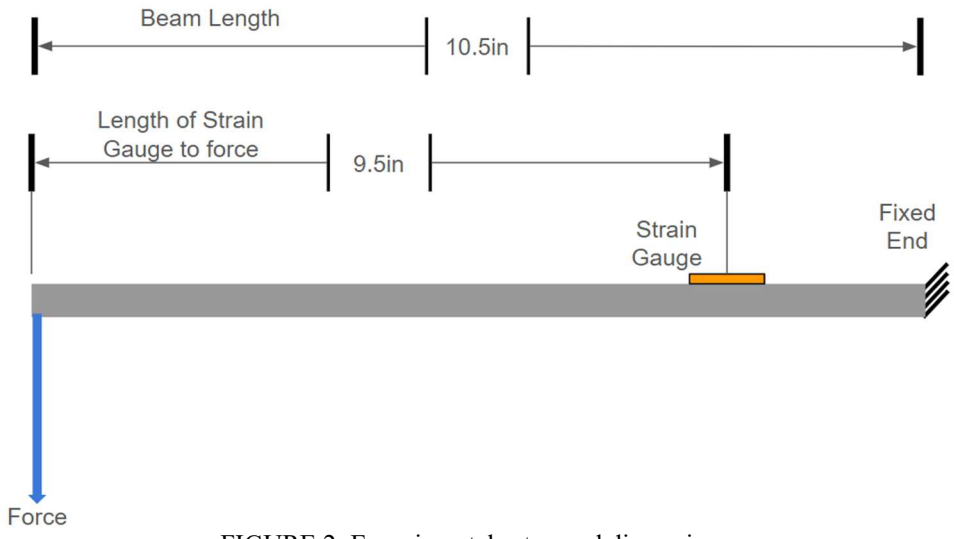


FIGURE 2: Experimental setup and dimensions.

Deliverable 2:

The experimental data was collected from the strain gauge for force values of zero to ten pounds. The force data was used to calculate the stress on the beam. This was done using equation 1 below.

$$\sigma = \frac{M * y}{I} \tag{8}$$

Where M is the moment, y is the distance to the centroid of the beam along the cross section of the face of interest, and I is the moment of inertia. The equation for the moment is calculated using the force, and the distance the force occurs from the base of the beam.

$$M = F * d \tag{9}$$

Due to the beam being symmetrical across its x and y axis, the y value can be simplified into the equation below.

$$y = \frac{t}{2} = 0.125in \tag{10}$$

The final variable needed is the moment of inertia, which is solved using the known dimensions of the beam, and represents the tendency of a shape to resist deformation and bending.

$$I = \frac{bt^3}{12} = 0.001302083in^4 \tag{11}$$

Using these equations, the stress on the beam is recorded. Now, the experimental values for strain are plotted along the stress for all eleven values. The resulting plot is below as FIG. 4.

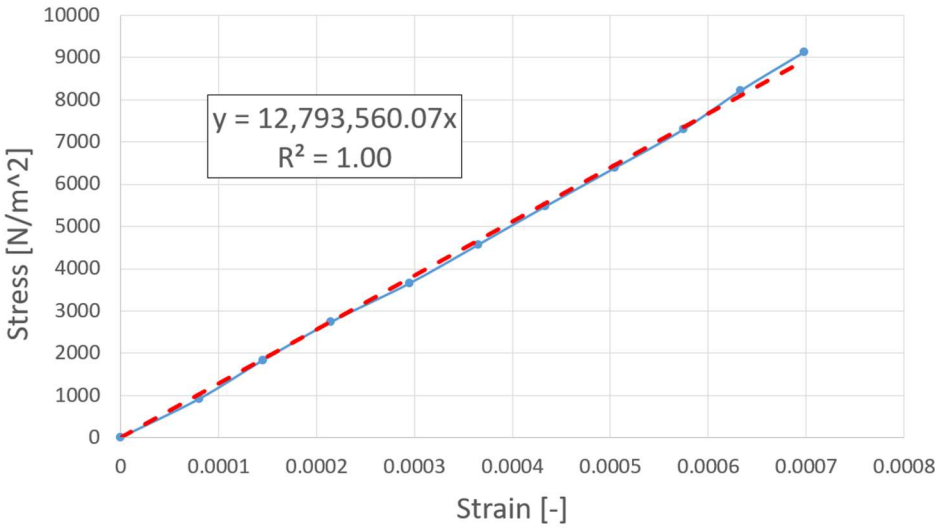


FIGURE 4: Experimental stress strain plot

FIG. 3 is linear, and because the material is metal (6061 Aluminum), this represents the elastic region of the stress strain plot. This means that using Hooke's law (equation 12), which states that there is a linear relationship between the stress and strain, the modulus of elasticity can be approximated to be the trendline.

$$E = \frac{\sigma}{\varepsilon} \tag{12}$$

This value is reported with a high certainty due to the R^2 value of one. This indicates that the trendline is a good representation of the data. The reported modulus of elasticity is 12,793,560.07[lbs/in^2]. The ideal value is 10,000,000[lbs/in^2]. The percent difference is 27.935% between these values.

Deliverable 3:

In order to properly support the experimental results, and add confidence to the results, a finite element analysis (FEA) in SolidWorks was completed on a simulated beam. The maximum deflection from the FEA will be compared to a calculated maximum deflection. This will give an idea of how the simulation compares to the calculated values.

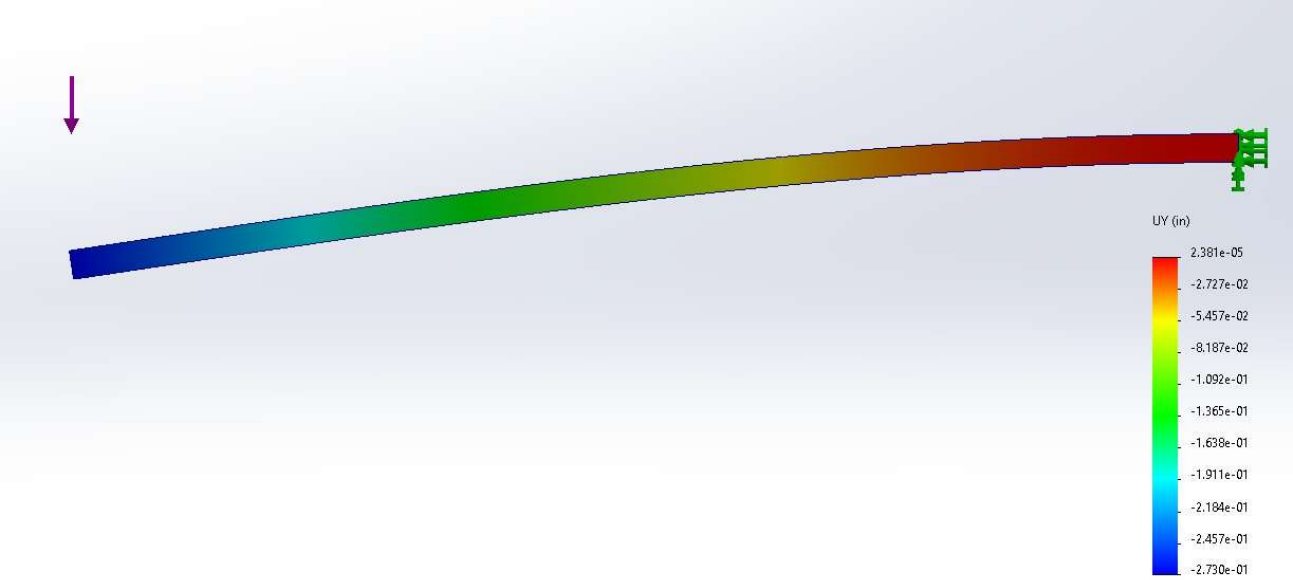


FIGURE 5: Side view of FEA beam deflection

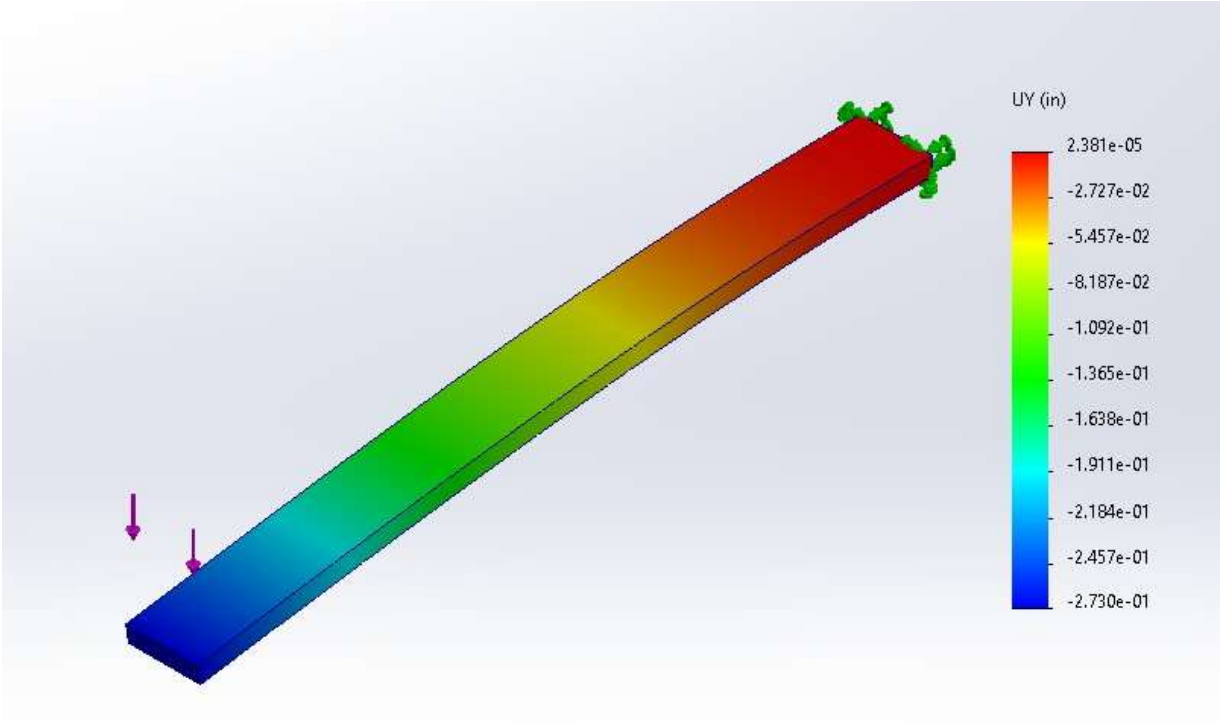


FIGURE 6: Isometric view of FEA beam deflection

The FEA analysis calculates a maximum deflection of 0.273in at the unsupported end of the beam.

To calculate the maximum deflection, the previously obtained values of maximum force, length of beam, the experimental modulus of elasticity, and the moment of inertia will be used in equation 13. These values are related through the maximum deflection equation below.

$$d_{max} = \frac{F_{max} * L_2^3}{3E * I} \tag{13}$$

Plugging in the values from deliverable 2,

$$d_{max} = \frac{10lb * (10.5in)^3}{3(12{,}793{,}560.07\text{Pa}) * 0.001302083in^4} = 0.21548654in \quad (13a)$$

Yielding a value of 0.21548654in, There is a 26.69004727% difference between the deflection of the simulation, and the calculated value.

Major talking point (1):

The experimental deflection is 0.215in, and the simulated value is 0.273in. This is a difference of 0.058in, with a percentage difference being 26.69004727%. This difference closely aligns with the difference between the ideal modulus of elasticity, and the measured modulus of elasticity which was 27.9356007%. The close relationship between the two hints that the error could stem from the difference in the modulus of elasticity.

Major talking point (2):

Due to the difference in modulus of elasticity, recomputing the maximum deflection calculations with the ideal modulus of elasticity of 10,000,000[lbs/in^2] can be done.

$$d_{max} = \frac{10lb * (10.5in)^3}{3(10{,}000{,}000\text{Pa}) * 0.001302083in^4} = 0.275684in \quad (13b)$$

This yields a value very similar to the FEA simulation, and the percentage difference drops to 0.973578445%. This result is much closer to the expected value and shows that a large difference in the resulting values stems from the difference in the modulus of elasticity used.

Deliverable 4:

One assumption made is constant properties. It is assumed that the 6061 aluminum has constant properties, when small differences occur between the same metals from different factories. This was shown in a 2019 study which tested nine samples of aluminum 6061 from nine manufacturers [1]. The resulting stress tests showed different levels of stress within the metal between samples. The mesh quality, and material differences lead the finite element analysis to not directly reflect real world data and can account for some discrepancy.

Another assumption made by finite element analysis to compute the maximum deflection is how it solves beams. First, it utilizes a mesh, where it layers the beam into triangles. Deflection calculations are completed at each of these triangles as it moves down the beam. This mesh is pictured below in FIG. 7.

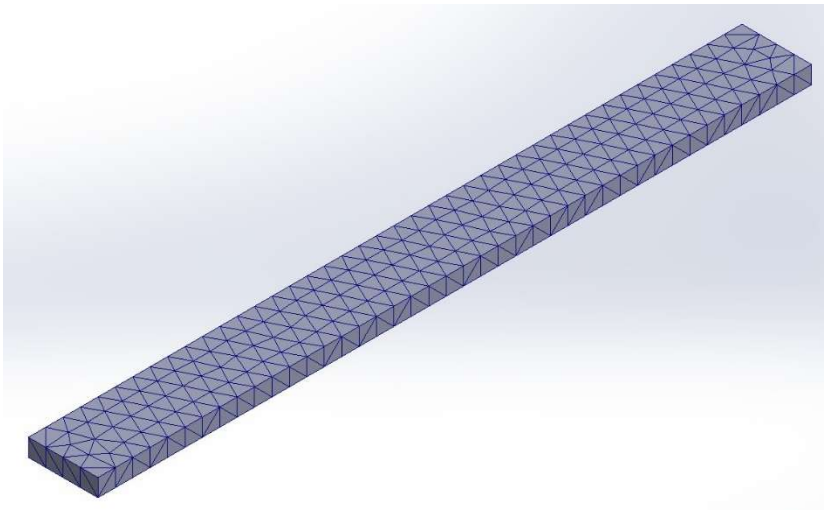


FIGURE 7: Mesh layer on beam

Finer meshes include more triangles, take longer to compute, and return results that better mirror real world examples. To show the importance of a finer mesh grid, a simulation will be run on the same beam using the roughest and finest mesh settings.

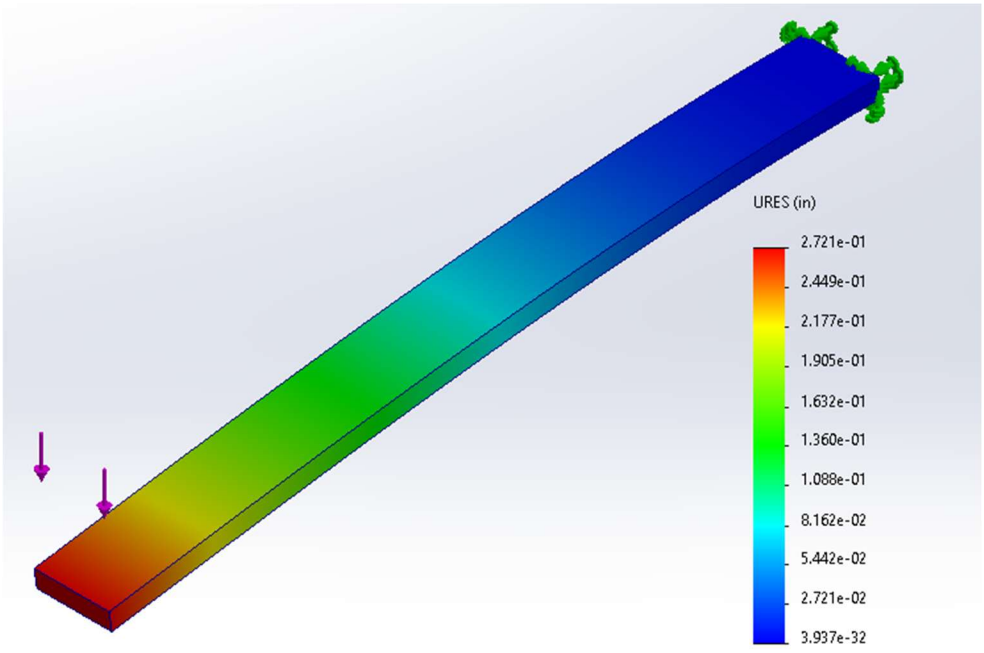


FIGURE 8: Rough mesh finite element analysis

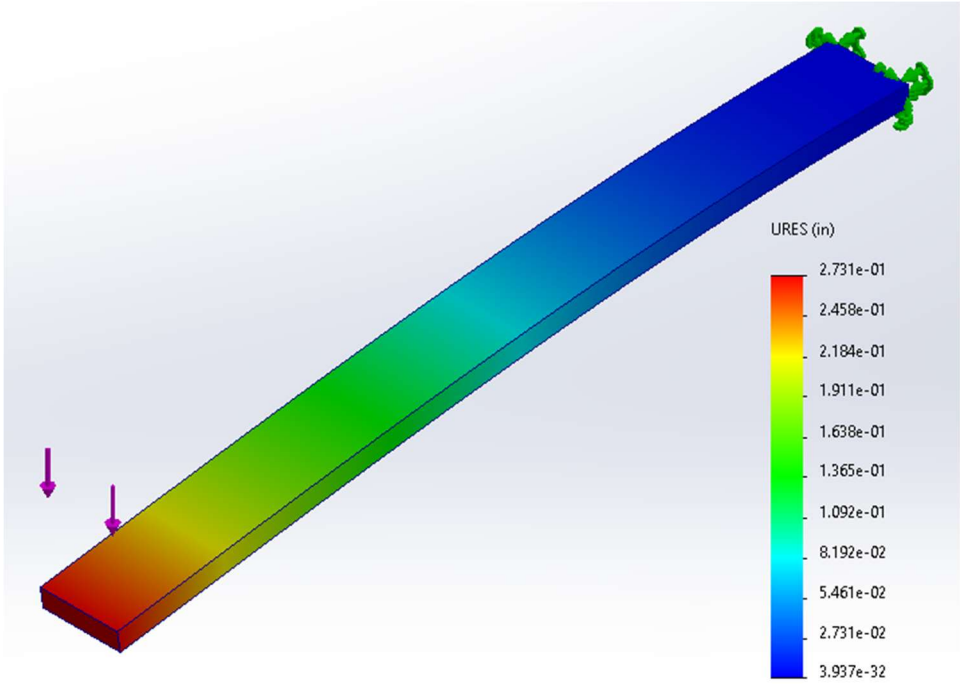


FIGURE 9: Fine mesh finite element analysis

The difference in displacement between the mesh in FIG. 8, and FIG. 9, is 0.01in of displacement. This is not a large value, but the finer mesh is closer to the recorded experimental values.

Boundary conditions are a part of finite element analysis, and also an assumption. They decide how the beam will behave, and finite element analysis assumes a perfectly constant and perpendicular force to the beam's original position, as well as a perfectly fixed end. As the deflection begins in the FEA software, the force is still constantly applied in the same direction. In reality, as the beam deflects, the force begins to be applied at an angle respective to the original beam position. This angle ensures that some force is lost along the beam, and no longer entirely perpendicular. This is not to be considered due to the small amount of deflection in respect to the length of the beam. The resulting angle is small, and therefore negligible force is lost. On the opposing side, as the beam deflects, the beam presses up against its support, but due to the relatively small force of 10lb, this loss is also negligible.

Constant temperature is an assumption from the FEA software, but it is not considered for this calculation. This is because the governing equation for the endurance limit, which has a temperature factor as k_d is as follows.

$$S_e = k_a k_b k_c k_d k_e S'_e \tag{14}$$

A higher temperature results in a lower temperature factor, which therefore decreases the endurance limit of the material, causing it to not last as many cycles. A temperature of 70°F returns a temperature factor of one. This means that there will be no effect on the endurance of the material, and therefore no effect on the strength.

REFERENCES

[1] B. S. Aakash, J. Connors, and M. D. Shields, “Stress-strain data for aluminum 6061-T651 from 9 lots at 6 temperatures under uniaxial and plane strain tension,” *Data in Brief*, vol. 25, p. 104085, Aug. 2019, doi: https://doi.org/10.1016/j.dib.2019.104085.

[2] Department of Mechanical Engineering, Clemson University. *ME 3330 – Mechanical Engineering Lab II Student Manual.* Fall 2025 ed., Clemson University, 2025.

[3]“Marin Endurance Limit Equations and Calculator,” *www.engineersedge.com*. https://www.engineersedge.com/calculators/marin_endurance_limit_15637.htm

[4] N Hale, *Strain Gauge Lectures Combined,* Clemson University 2026,

[5]“Strain Gauge: What Is It? How Is It Used? Types, Application,” *www.iqsdirectory.com*. https://www.iqsdirectory.com/articles/load-cell/strain-gauge.html

[6]“What is a Wheatstone Bridge?,” *https://www.dwyeromega.com/en-us/*, 2025. https://www.dwyeromega.com/en-us/resources/wheatstone-bridge?srsltid=AfmBOoodIMemGdYOwvGjwU0xiLQlcdTSzvlIm5UseTEibaHU0nEAHJ-a (accessed Apr. 14, 2026).